IEEE SIGHT · SDG SPRINTS 2026

SOLUTION CONCEPT DECK · CHECKPOINT 3

AgriLoop

A proposed business with a free donation channel, helping retail and wholesale vendors offer eligible surplus for discounted sale or free donation, collected by the buyer or recipient.

"Small acts, when multiplied by millions of people, can transform the world" – Howard Zinn –

Team name
Fade

University
General Sir John Kotelawala Defence University

Problem statement ID
[2.2], [12.2], [1.3] (confirm)

SDG(s) addressed
SDG 12 (target 12.3) and SDG 2 (target 2.1)

Submission no.
1

Team leader
Mohanaram Uvaram

Deck updated
27.10.2026

Team & mentors

Mohanaram Uvaram
Team leader
Backend, reservation logic, testing
KDU · Software Engineering · Year 2

Manidu Samaranayake
Team member
Interface and usability testing
KDU · Software Engineering · Year 2

Yasith Peiris
Team member
Coordination, vendor interviews, presentation
KDU · Software Engineering · Year 2

All three members
Field notes, evidence review and final decisions.

University / programme
KDU, Software Engineering. All three members are in their second year.

Mentor / supervisor
No mentor yet. We are seeking a food-safety advisor to guide a later live-food pilot.

The problem and our preliminary evidence

Problem statement (preliminary)

Fade's earlier submission reported unsold food near closing time and unsold produce at wholesale markets in Colombo and nearby areas.

A new documented study is needed to measure how much eligible surplus could reach buyers at an acceptable total cost.

32.8%
Sri Lankan households food insecure*
BRIGHT 2024–25 [1]

35.3%
urban households food insecure*
BRIGHT 2024–25 [1]

~42%
Colombo waste: restaurants, markets & meat shops*
Colombo study [2]

Who is affected, where, how many

- Retail buyers near bakeries and cafeterias.
- Wholesale kitchens, canteens and vendors with rejected produce (Peliyagoda: earlier lead, unconfirmed).
- Proposed outreach areas: Wellawatte, Ratmalana, Panadura. No vendors or recipients recruited yet.

Root causes (working hypotheses, to be tested)

- **Targeted:** arranging buyers at closing may cost more than the surplus recovers.
- Some food has a short usable window.
- Buyers and recipients may not reach pickup in time.

*** Historical context, not October 2026 and not our market.** 32.8% of Sri Lankan households experienced moderate or severe food insecurity during the 12 months preceding the 2024–25 BRIGHT survey; 35.3% of urban households, using the same measure (BRIGHT 2024–25, Table 2; 12-month recall) [1]. The ~42% share comes from restaurants, markets and meat shops in earlier Colombo research cited by [2]; it is not recoverable supply.

Who we are designing for

Primary user

- **Retail:** nearby buyers collecting small bundles (proposed outreach: Wellawatte, Ratmalana, Panadura).
- **Wholesale:** kitchens, canteens and welfare homes collecting kilogram lots with their own transport.
- **Donations:** proposed policy: approved organisations first; individual access remains an open decision.
- Age, income and phone access: not established.

Current practice (reported earlier)

- **Vendors:** discount, donate, give to staff or discard at closing.
- **Wholesalers:** rejected or unsold produce has no quick buyer.
- **Buyers:** cannot see what is left or reach the vendor in time.

Reported in the earlier submission; not independently verified.

Other stakeholders

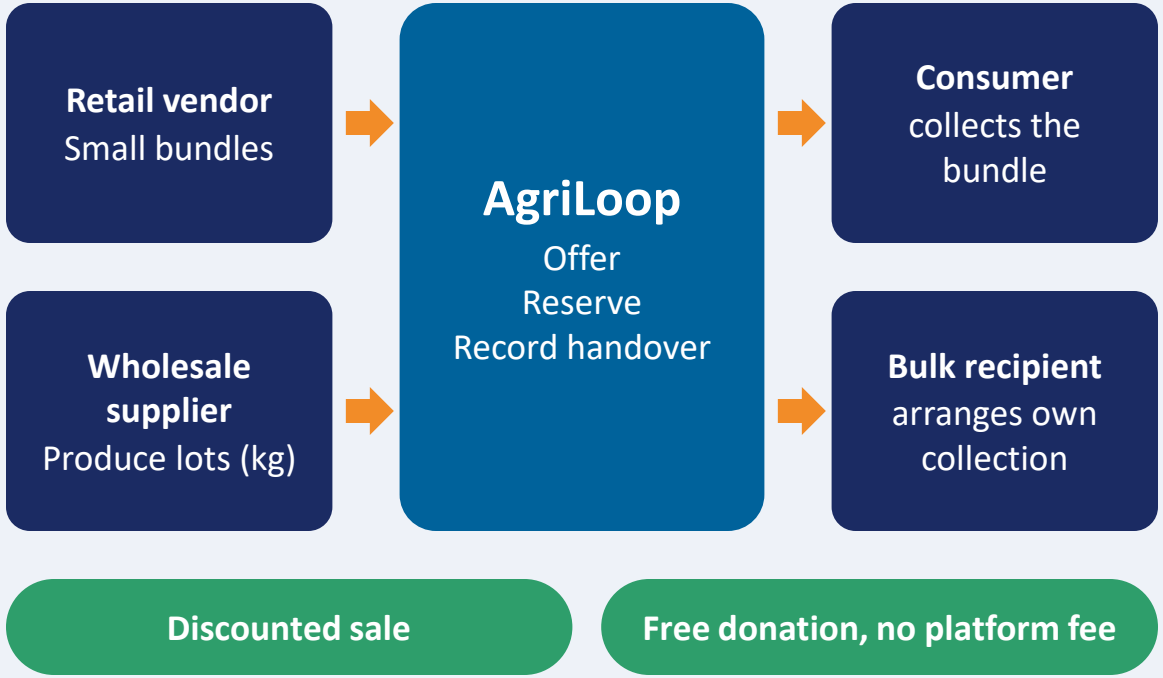
- **Pays:** buyers (sales); vendors if a fee is chosen.
- **Enables:** vendors, bulk recipients, a food-safety advisor.
- **Could block:** regulators, and vendors unwilling to list.
- Vendors and recipients not yet recruited; donation funding not secured.

User evidence: preliminary, unverified

The earlier submission reported a 42-person survey and preliminary stakeholder interviews. Original responses and interview records are unavailable for verification. These findings are preliminary and require a new documented study.

Our solution

Concept diagram; nothing is built or tested yet



Buyer or recipient collects. AgriLoop provides no delivery.

One-sentence solution (proposed)

We help food vendors move eligible surplus to nearby buyers and recipients by listing it for discounted sale or free donation, collected by the buyer or recipient.

1

Two streams, one platform · Planned

Retail bundles and wholesale lots share accounts and records.

2

Sale or free donation · Planned

Donations carry no food price and no platform fee.

3

Reserve, collect, record · Planned

Stock-controlled reservations; the vendor confirms the actual handover.

How it works

1 · Input

Vendor lists eligible surplus: stream, unit, quantity, price or donation, pickup window.

2 · Process

Server checks policy and stock. Buyer reserves; a bulk request waits for vendor acceptance.

3 · Output

Reservation reference (QR optional), pickup instructions and an updated stock count.

4 · User action

Collect in the window, check the food; vendor confirms the actual handover.

Technology & components (proposals)

- **Idea stage:** nothing built or tested yet. All Planned.
- Functional web app: accounts, listings, reservations and handover records; simulated food data.
- Server rules for stock and permissions; Supabase/PostgreSQL proposed.
- Admin console: vendor review, suspension, incidents, audit trail.
- Costed option: Render hosting and Supabase. Paid setup depends on funding.

User experience

- Vendor posts from a template and confirms quantity and time.
- Buyer browses a simple list (map optional) and sees total price and pickup rules before reserving.
- Bulk recipient confirms capacity and arranges transport; accepted weight is recorded (illustrative: 27 of 30 kg).
- Sinhala, Tamil and English are language goals, to be tested.

What exists, and what is new

Solution	Sales and donations	Bulk produce and collection	Transfer records
Too Good To Go [3][4]	Sells Surprise Bags at half price or less; donation workflow not established	Bags: in-store pickup at a set time. Parcels: manufacturer and wholesale goods, delivered	Pick-up Code redeemed at the store [3]
Robin Hood Army [5]	Volunteer redistribution of surplus food; zero-funds organisation	Volunteer-led distribution; Sri Lanka listed in its global network. Bulk produce not established	Not established from reviewed sources
AgriLoop (proposed)	**Discounted sale or free donation in both streams**	**Retail bundles and bulk lots; buyer or recipient collects; no delivery**	**Reservation and handover records, actual weight**

Innovation type (tick one)

☐ Existing innovation introduced to Sri Lanka
☒ **Existing innovation adapted to Sri Lankan conditions**
☐ Improvement to an existing Sri Lankan solution
☐ New invention

Where we searched

Reviewed 25–26 Oct 2026: official Too Good To Go and Robin Hood Army sources. Planned: patent and academic searches.

Our combination is a proposed adaptation to test.

Impact & SDG alignment

SDG 12

Primary
Responsible consumption and production

SDG 2

Secondary
Zero hunger

SDG target & indicator

- Target 12.3: food waste and losses; indicator 12.3.1, food loss and waste indices [8].
- Target 2.1: access to food; indicator 2.1.2, moderate or severe food insecurity [9].
- Our kg and savings would be project measures; they do not replace these indicators.

Live-pilot measure (later)	Baseline	Target	Measured by
Food collected (kg, portions separately)	Not measured	Set after baseline	Handover records
Collection rate	Not measured	Set after baseline	Completed ÷ due
Net saving after travel	Not measured	Set after baseline	Comparable price − paid − travel

Initial four-week phase: prototype measures (proposed)

- All four workflows demonstrated: retail and wholesale, sale and donation.
- Tasks completed independently, with help or not at all.
- Price or free status and collection arrangements understood.
- Cancellation, expiry and partial wholesale acceptance tested.
- Test stock rules against double booking.

Simulated transfers never count as food rescued, meals or waste prevented. This test does not establish demand.

Is it realistic in Sri Lanka?

Technical feasibility

- Proposal: working web prototype, tested locally and hosted when funded.
- Risk: double booking, duplicate pickup.
- Test: stock-control tests in the prototype.

Cost & affordability

- Proposed cash budget: LKR 60,000 for build/test and two months online.
- Team: LKR 5,000. Funding gap: LKR 55,000.
- Existing laptops and phones; labour unpaid.
- Estimate, not a quote or launch budget.

Materials & local supply

- No custom hardware planned; phones and internet needed.
- Proposed outreach: Wellawatte, Ratmalana, Panadura; Peliyagoda is an unconfirmed earlier lead.
- Vendors and recipients not yet recruited.

Regulation & safety

- Food (Hygiene) Regulations 2011 apply to food handling [6].
- No real food in this phase; no approval or readiness claimed.
- Safety advisor needed before any live pilot.

Adoption by users

- To test: can vendors list at closing and buyers collect?
- Willingness is not commitment; no demand is established.
- Simulated listings test usability only.

Sustainability & model

- A proposed business with a free donation channel.
- Fee or subscription undecided; no sponsor secured.
- Donation costs tracked separately.
- Base hosting after month 2: about LKR 19,950/month, before extras [7][10].

Functional prototype plan & budget

Week 1 · Proposed

Approach participants, document interviews, define workflows, start surplus observations where access is granted.

Week 2 · Proposed

Create listings and reservation workflows for both streams.

Week 3 · Proposed

Add donation handling, bulk acceptance, handover records, cancellation scenarios.

Week 4 · Proposed

Run participant sessions and technical tests, fix problems, document findings.

Current stage (select one): ☒ Idea ☐ Concept design ☐ Early prototype ☐ Functional prototype ☐ Tested with users

Prototype scope · Planned

Will build: working accounts, listings, reservations and handover records for both streams and modes. Simulated food data.

Recruit (targets, subject to access): 2 retail vendors, 1 wholesale vendor, 4 potential buyers, 1 bulk-recipient representative.

Observe: five trading days of surplus logs, including zero-surplus days. No redistribution.

Excluded: real food, delivery, payments, native mobile apps and hired developers.

Four-week build; two months hosting. Labour/devices in-kind. Gap: LKR 55,000.

Proposed cash budget [7][10]	LKR
Hosting + team workspace (2 mo.)	22,400
Database/auth/backups (2 mo.)	17,500
Domain (1 year allowance)	5,000
Travel, data and printing	4,000
Contingency / taxes / overages	11,100
Total estimate	**60,000**

Risks, ethics & inclusion

Risk	Likelihood	Impact	Mitigation
Technical: double booking, duplicate pickup	Medium	Medium	Plan atomic stock updates and test concurrent bookings and duplicate handovers.
Recruitment not secured	High	High	Report the limitation; keep wholesale and recipient workflows as simulated demonstrations.
Funding gap: LKR 55,000	High	High	Develop locally; approve paid services only when funded.
Food safety and regulation	Not assessed	High	No real food in this phase; advisor-reviewed rules before any later pilot.

Key operational assumption for a later live pilot

Surplus may vary by day. Log quantities and zero-surplus days. Test whether vendors can list in time and recipients can collect affordably in a later pilot. Recruitment is not secured.

Ethics, inclusion & environment

- **Privacy:** minimal data; contact details private.
- **Safety:** vendors can withdraw food; collectors can reject it.
- **Access:** Sinhala, Tamil, English goals; list view; no hardship labels.
- **Containers: assign responsibility before a live pilot.**

Before a later real-food pilot

Confirm applicable requirements, safety advice, eligible categories, handling, collection, incident procedures and a separate budget. No approval or readiness is claimed.

11 · CLOSING · ~25 s

AgriLoop aims to turn eligible surplus into discounted sales and free donations.

The problem

Unsold food and produce were reported in an earlier submission. 32.8% of Sri Lankan households experienced moderate or severe food insecurity in the 12 months preceding the 2024–25 BRIGHT survey [1].

Our solution

A proposed platform for retail and wholesale surplus, with discounted sales or free donations collected by buyers or recipients.

Our ask

Seeking LKR 55,000 funding, vendor and recipient access for testing, and safety advice for a later live-food pilot.

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References

References (IEEE format)

[1] D. Headey et al., “Food insecurity in post-crisis Sri Lanka: Evidence from the 2024–2025 BRIGHT survey,” IFPRI, 2025, Table 2. Accessed: Oct. 18, 2026. [Online]. Available: https://cgspace.cgiar.org/server/api/core/bitstreams/3714264a-7860-4a77-b9dc-1dffb100e327/content

[2] M. Reitemeier, M. Aheeyar and P. Drechsel, “Perceptions of food waste reduction in Sri Lanka’s commercial capital, Colombo,” Sustainability, vol. 13, no. 2, art. 838, 2021, doi: 10.3390/su13020838.

[3] Too Good To Go, “Marketplace Store Terms.” Accessed: Oct. 25, 2026. [Online]. Available: https://www.toogoodtogo.com/en-us/legal/terms-and-conditions-marketplace

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[5] Robin Hood Army, “The Idea” and “Global Presence.” Accessed: Oct. 26, 2026. [Online]. Available: https://robinhoodarmy.com/

[6] Sri Lanka, Food (Hygiene) Regulations, 2011, Gazette Extraordinary No. 1742/26, Jan. 26, 2012. Accessed: Oct. 25, 2026. [Online]. Available: https://leap.unep.org/en/countries/lk/national-legislation/food-hygiene-regulations-2011

[7] Render, “Pricing” and workspace plans. Accessed: Oct. 26, 2026. https://render.com/pricing ; https://render.com/docs/platform-features-by-plan

[8] United Nations, “Goal 12: Ensure sustainable consumption and production patterns” (target 12.3, indicator 12.3.1). [Online]. Available: https://sdgs.un.org/goals/goal12

[9] United Nations, “Goal 2: Zero hunger” (target 2.1, indicator 2.1.2). https://sdgs.un.org/goals/goal2

[10] Supabase, “Pricing.” Accessed: Oct. 26, 2026. https://supabase.com/pricing

Preliminary evidence and open decisions

Preliminary evidence: unverified

The earlier submission reported a 42-person survey and preliminary stakeholder interviews. Original responses and interview records are unavailable for verification. These findings are preliminary and require a new documented study.

Reported in the earlier submission; not independently verified: 4 market vendors, 3 bakery or cafeteria managers, 2 commission agents, 1 driver and 1 kitchen coordinator were interviewed; 15–30 unsold bakery items; 40–70 kg/day of market discards. These are items and kilograms, not meals, and not all is eligible for redistribution.

Willingness expressed in interviews is not a partnership, commitment or proven demand.

Secondary sources used

BRIGHT survey [1]: Table 2, 12-month recall (the report text prints 35.5% urban; we use the table value, 35.3%). Colombo study [2]: the ~42% share is cited there from earlier research.

Open decisions

- First sites and food categories
- Donations: individual access (proposed: organisations first)
- Payment, fee or subscription model
- LKR 55,000 funding gap; final stack; later pilot
- FoodLoop: identity and features need verification